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Probability and Statistics

Problem Set 1

Exercise 1:

An urn contains five red balls numbered: 0, 0, 0, 1, 1, three green balls numbered 0, 1, 2, and two black balls numbered 0, 2.

I) In a successive draw without replacement of three balls, in how many ways can we draw:

1. $\Omega =$ (Three balls).
2. $A =$ (Three balls of the same color).
3. $B =$ (Three balls of different colors pairwise).
4. $C =$ (Exactly two red balls).
5. $D =$ (At least one green ball).
6. $E =$ (At most two red balls).
7. $F =$ (Three balls bearing the same number).
8. $G =$ (Three balls of the same color and bearing the same number).
9. $H =$ (Three balls of different colors pairwise and bearing the same number).
10. $I =$ (Exactly two balls of the same color bearing odd numbers).
11. $J =$ (Three balls bearing numbers whose product is zero).

II) Redo the same calculations for a successive draw with replacement.

Correction

Total balls: red (0,0,0,1,1) + green (0,1,2) + black (0,2) = 10 balls

1. $\Omega = C_{10}^3 = 120$
2. $A =$
   - All red: $C_5^3 = 10$
   - All green: $C_3^3 = 1$
   - All black: $C_2^3 = 0$

$P(A) = 10 + 1 = 11$

3. $B =$ (One of each color) $C_5^1 \times C_3^1 \times C_2^1 = 5 \times 3 \times 2 = 30$; $P(B) = 30$

4. $C = C_5^2 \times C_5^1 = 10$; $P(C) = 10 \times 5 = 50$

5. $D =$ (At least one green ball):

   - Total ways: $C_{10}^3 = 120$
   - No green: $C_7^3 = 35$

   $P(D) = 120 - 35 = 85$

6. $E =$ (At most two red balls):

   - Total: 120
   - Three red: $C_5^3 = 10$

   $P(E) = 120 - 10 = 110$

7. $F =$ (Three balls bearing the same number):

   - All 0's: $C_6^3 = 20$
   - All 1's: $C_2^3 = 0$
   - All 2's: $C_2^3 = 0$

   $P(F) = 20$

8. $G =$ (Three balls of the same color and bearing the same number):

   - Red-0: $C_3^3 = 1$
   - Others: 0 (not enough balls)

   $P(G) = 1$

9. $H =$ (Three balls of different colors pairwise and bearing the same number):

   - All 0's: $C_3^1 \times C_1^1 \times C_2^1 = 3 \times 1 \times 2 = 6$ (red-0, green-0, black-0)
   - Other numbers: not possible

   $P(H) = 6$

10. $I =$ (Exactly two balls of the same color bearing odd numbers):

    - Odd numbers available: only 1 (red-1, green-1)
    - Choose color with 2 odd balls: only red works (has two 1's)
    - Choose 2 red-1: $C_2^2 = 1$
    - Choose 1 other ball (not red-1): $C_8^1 = 8$

    $P(I) = 1 \times 8 = 8$

11. $J =$ (Three balls bearing numbers whose product is zero):
    - Total: 120
    - Product non-zero: all balls must have non-zero numbers
    - Non-zero numbers: red-1 (2 balls), green-1,2 (2 balls), black-2 (1 ball) = 5 balls
    - Ways with product non-zero: $C_5^3 = 10$

    $P(J) = 120 - 10 = 110$

II) Successive draw with replacement

1. Ω: $10^3 = 1000$
2. $A =$ (Three balls of the same color):
    - All red: $5^3 = 125$
    - All green: $3^3 = 27$
    - All black: $2^3 = 8$

    $P(A) = 125 + 27 + 8 = 160$
3. $B =$ (Three balls of different colors pairwise):
    - Permutations of colors: $3! = 6$
    - Red, green, black: $5 \times 3 \times 2 = 30$

    $P(B) = 6 \times 30 = 180$
4. $C =$ (Exactly two red balls):
    - Choose positions for red: $C_3^2 = 3$
    - Two red: $5^2 = 25$
    - One non-red: $5^1 = 5$

    $P(C) = 3 \times 25 \times 5 = 375$
5. $D =$ (At least one green ball):
    - Total: 1000
    - No green: $7^3 = 343$

    $P(D) = 1000 - 343 = 657$
6. $E =$ (At most two red balls):
    - Total: 1000
    - Three red: $5^3 = 125$

    $P(E) = 1000 - 125 = 875$

7. $F =$ (Three balls bearing the same number):

   - Number 0: $6^3 = 216$
   - Number 1: $3^3 = 27$
   - Number 2: $3^3 = 27$

   $P(F) = 216 + 27 + 27 = 270$

8. $G =$ (Three balls of the same color and bearing the same number):

   - Red-0: $3^3 = 27$
   - Red-1: $2^3 = 8$
   - Green-0: $1^3 = 1$, Green-1: $1^3 = 1$, Green-2: $1^3 = 1$
   - Black-0: $1^3 = 1$, Black-2: $1^3 = 1$

   $P(G) = 27 + 8 + 1 + 1 + 1 + 1 + 1 = 40$

9. $H =$ (Three balls of different colors pairwise and bearing the same number):

   - Only possible with number 0
   - Permutations: $3! = 6$
   - Red-0, green-0, black-0: $3 \times 1 \times 1 = 3$

   $P(H) = 6 \times 3 = 18$

10. $I =$ (Exactly two balls of the same color bearing odd numbers):

    - Only odd number is 1
    - Choose color with 2 odd balls: only red works
    - Choose positions for red-1: $C_3^2 = 3$
    - Two red-1: $2^2 = 4$
    - One other ball (not red-1): $8^1 = 8$

    $P(I) = 3 \times 4 \times 8 = 96$

11. $J =$ (Three balls bearing numbers whose product is zero):

    - Total: 1000
    - Product non-zero: all balls must have non-zero numbers
    - Non-zero numbers: 5 balls (red-1:2, green-1,2:2, black-2:1)
    - Ways with product non-zero: $5^3 = 125$

    $P(J) = 1000 - 125 = 875$

Exercise 2:

An urn (A) contains five black balls, two green balls, and one red ball. Another urn (B) contains two black balls and one green ball. We simultaneously draw two balls from urn (A) and one ball from urn (B).

In how many ways can we draw:

1. Ω: (Three balls).
2. C: (Three green balls).
3. D: (Exactly two green balls).
4. E: (At least one green ball).
5. F: (At most one green ball).

Correction

Given:

- Urn A: 5 black (B), 2 green (G), 1 red (R) → Total: 8 balls
- Urn B: 2 black (B), 1 green (G) → Total: 3 balls
- Draw: 2 balls from A + 1 ball from B simultaneously

1. $\Omega =$ (Three balls): $C_8^2 \times C_3^1 = 28 \times 3 = 84$
2. $C =$ (Three green balls):
   - From A: need 2 green from 2 available: $C_2^2 = 1$
   - From B: need 1 green from 1 available: $C_1^1 = 1$

   $P(C) = 1 \times 1 = 1$
3. $D =$ (Exactly two green balls):

   Case 1: Both greens from A, non-green from B
   - A: 2 green from 2: $C_2^2 = 1$
   - B: 1 non-green from 2 (black only): $C_2^1 = 2$

   Ways: $1 \times 2 = 2$

   Case 2: One green from A, one green from B, one non-green from A
   - A: 1 green from 2 + 1 non-green from 6: $C_2^1 \times C_6^1 = 2 \times 6 = 12$
   - B: 1 green from 1: $C_1^1 = 1$

   Ways: $12 \times 1 = 12$

   Total $P(D) = 2 + 12 = 14$
4. $E =$ (At least one green ball):

- Total: 84
- No green: All balls non-green
- From A: 2 non-green from 6 (5 black + 1 red): $C_6^2 = 15$
- From B: 1 non-green from 2 (black only): $C_2^1 = 2$

No green ways: $15 \times 2 = 30$

$P(E) = 84 - 30 = 54$

5. $F =$ (At most one green ball):

Case 1: No green balls (calculated above): 30

Case 2: Exactly one green ball:

Subcase 2a: Green from A only

- A: 1 green from 2 + 1 non-green from 6: $C_2^1 \times C_6^1 = 12$
- B: 1 non-green from 2: $C_2^1 = 2$

Ways: $12 \times 2 = 24$

Subcase 2b: Green from B only

- A: 2 non-green from 6: $C_6^2 = 15$
- B: 1 green from 1: $C_1^1 = 1$

Ways: $15 \times 1 = 15$

Exactly one green: $24 + 15 = 39$

Total $P(F) = 30 + 39 = 69$

Exercise 3:

1) Calculate:

$$S_0 = \sum_{k=0}^{n} C_n^k, \quad S_1 = \sum_{k=0}^{n} k C_n^k,$$

$$S_2 = \sum_{k=0}^{n} k(k-1) C_n^k, \quad S_2' = \sum_{k=0}^{n} k^2 C_n^k$$

2) Verify:

$$\sum_{y=0}^{k} C_a^y C_b^{k-y} = C_{a+b}^k,$$

$$\sum_{y=0}^{n} (C_n^y)^2 = C_{2n}^n,$$

$$C_n^p = C_n^{n-p},$$

$$C_{n-1}^{p-1} = \frac{p}{n} C_n^p$$

Correction

1) Calculate the sums:
We use the binomial theorem: $(1+x)^n = \sum_{k=0}^{n} C_n^k x^k$
(a) $S_0 = \sum_{k=0}^{n} C_n^k$

$$\text{Take } x = 1 : \quad (1+1)^n = \sum_{k=0}^{n} C_n^k (1)^k$$

$$S_0 = 2^n$$

(b) $S_1 = \sum_{k=0}^{n} k C_n^k$
Differentiate the binomial expansion:

$$\frac{d}{dx}(1+x)^n = \frac{d}{dx}\left(\sum_{k=0}^{n} C_n^k x^k\right)$$

$$n(1+x)^{n-1} = \sum_{k=0}^{n} k C_n^k x^{k-1}$$

Multiply both sides by x:

$$nx(1+x)^{n-1} = \sum_{k=0}^{n} k C_n^k x^k$$

Take $x = 1$:

$$n \cdot 1 \cdot (1+1)^{n-1} = \sum_{k=0}^{n} k C_n^k$$

$$S_1 = n \cdot 2^{n-1}$$

(c) $S_2 = \sum_{k=0}^{n} k(k-1) C_n^k$
Differentiate twice:

$$\frac{d^2}{dx^2}(1+x)^n = \frac{d^2}{dx^2}\left(\sum_{k=0}^{n} C_n^k x^k\right)$$

$$n(n-1)(1+x)^{n-2} = \sum_{k=0}^{n} k(k-1) C_n^k x^{k-2}$$

Multiply both sides by x^2:

$$n(n-1)x^2(1+x)^{n-2} = \sum_{k=0}^{n} k(k-1) C_n^k x^k$$

Take $x = 1$:

$$n(n-1) \cdot 1^2 \cdot (1+1)^{n-2} = \sum_{k=0}^{n} k(k-1) C_n^k$$

$$S_2 = n(n-1) \cdot 2^{n-2}$$

(d) $S_2' = \sum_{k=0}^{n} k^2 C_n^k$

We have: $k^2 = k(k-1) + k$

$$S_2' = \sum_{k=0}^{n} k^2 C_n^k = \sum_{k=0}^{n} [k(k-1) + k] C_n^k$$

$$S_2' = \sum_{k=0}^{n} k(k-1) C_n^k + \sum_{k=0}^{n} k C_n^k$$

$$S_2' = S_2 + S_1 = n(n-1)2^{n-2} + n2^{n-1}$$

$$S_2' = n2^{n-2}[(n-1) + 2] = n(n+1)2^{n-2}$$

2) Verify the identities:
(a) $\sum_{y=0}^{k} C_a^y C_b^{k-y} = C_{a+b}^k$
This is Vandermonde's identity.
(b) $\sum_{y=0}^{n} (C_n^y)^2 = C_{2n}^n$
Special case of Vandermonde with $a = b = n$ and $k = n$:

$$\sum_{y=0}^{n} C_n^y C_n^{n-y} = C_{2n}^n$$

But $C_n^{n-y} = C_n^y$, so:

$$\sum_{y=0}^{n} (C_n^y)^2 = C_{2n}^n$$

(c) $C_n^p = C_n^{n-p}$
Algebraic proof:

$$C_n^{n-p} = \frac{n!}{(n-p)!(n-(n-p))!} = \frac{n!}{(n-p)!p!} = C_n^p$$

(d) $C_{n-1}^{p-1} = \frac{p}{n} C_n^p$
Algebraic proof:

$$C_{n-1}^{p-1} = \frac{(n-1)!}{(p-1)!(n-p)!}$$

$$C_n^p = \frac{n!}{p!(n-p)!} = \frac{n}{p} \cdot \frac{(n-1)!}{(p-1)!(n-p)!}$$

Therefore:

$$C_{n-1}^{p-1} = \frac{p}{n} C_n^p$$

Exercise 4:

An elevator serving N floors contains S people.

1. In how many ways can the S people stop at these floors?
2. Suppose the S people stop as follows: there are n_i floors such that at each of them a_i people stop ($1 \leq i \leq k$) with $n_1 + \ldots + n_k = N$. In how many ways is this

possible?

Correction

1) In how many ways can the S people stop at these floors?

$$\text{Number of ways} = N^S$$

2) Suppose the S people stop as follows: there are n_i floors such that at each of them a_i people stop $(1 \leq i \leq k)$ with $n_1 + \ldots + n_k = N$. In how many ways is this possible?

Step 1: Choose which floors get which number of people The number of ways to assign the floor types is:

$$\frac{N!}{n_1!n_2!\cdots n_k!}$$

Step 2: Distribute people among the chosen floors The number of ways to partition the people is:

$$\frac{S!}{(n_1a_1)!(n_2a_2)!\cdots(n_ka_k)!}$$

Step 3: Distribute people within each floor type The number of ways to do this for type i is:

$$\frac{(n_ia_i)!}{(a_i!)^{n_i}}$$

Final Answer: Multiplying all these factors together:

$$\text{Number of ways} = \frac{N!}{n_1!n_2!\cdots n_k!} \times \frac{S!}{(n_1a_1)!(n_2a_2)!\cdots(n_ka_k)!} \times \prod_{i=1}^{k} \frac{(n_ia_i)!}{(a_i!)^{n_i}}$$

Simplifying:

$$\text{Number of ways} = S! \times \frac{N!}{\prod_{i=1}^{k} n_i!(a_i!)^{n_i}}$$

Exercise 5:

Let $(\Omega, \mathcal{A}, P)$ be a probability space, A, B, C elements of $\mathcal{A}$.

1. Show that $P(A \cap B) \geq P(A) - P(\overline{B})$.
2. Deduce that $P(A \cap B \cap C) \geq 1 - P(A) - P(B) - P(C)$.
3. Generalize to n events $A_1, \ldots, A_n$.

Correction

1) Show that $P(A \cap B) \geq P(A) - P(\overline{B})$
We start with the basic probability identity:

$$P(A) = P(A \cap B) + P(A \cap \overline{B})$$

Since $P(A \cap \overline{B}) \leq P(\overline{B})$ (because $A \cap \overline{B} \subseteq \overline{B}$), we have:

$$P(A \cap B) \geq P(A) - P(\overline{B})$$

2) Deduce that $P(A \cap B \cap C) \geq 1 - P(\overline{A}) - P(\overline{B}) - P(\overline{C})$
We apply the result from part 1 multiple times. First, apply it to events A and $B \cap C$:

$$P(A \cap (B \cap C)) \geq P(A) - P(\overline{B \cap C})$$

But $\overline{B \cap C} = \overline{B} \cup \overline{C}$, so:

$$P(\overline{B \cap C}) = P(\overline{B} \cup \overline{C}) \leq P(\overline{B}) + P(\overline{C})$$

Therefore:

$$P(A \cap B \cap C) \geq P(A) - [P(\overline{B}) + P(\overline{C})]$$

$$P(A \cap B \cap C) \geq 1 - P(\overline{A}) - P(\overline{B}) - P(\overline{C})$$

3) Generalize to n events $A_1, \ldots, A_n$
For n events $A_1, A_2, \ldots, A_n$, we have:

$$P\left(\bigcap_{i=1}^{n} A_i\right) \geq 1 - \sum_{i=1}^{n} P(\overline{A_i})$$

Using De Morgan's law and the union bound:

$$P\left(\bigcap_{i=1}^{n} A_i\right) = 1 - P\left(\overline{\bigcap_{i=1}^{n} A_i}\right) = 1 - P\left(\bigcup_{i=1}^{n} \overline{A_i}\right)$$

By the union bound (Boole's inequality):

$$P\left(\bigcup_{i=1}^{n} \overline{A_i}\right) \leq \sum_{i=1}^{n} P(\overline{A_i})$$

Therefore:

$$P\left(\bigcap_{i=1}^{n} A_i\right) \geq 1 - \sum_{i=1}^{n} P(\overline{A_i})$$

This inequality is known as the **Bonferroni inequality** and is useful in probability theory and statistics, particularly in multiple testing problems.

Exercise 6:

Show that:

The specification of a probability P on $(\Omega = \{e_1, \ldots, e_n, \ldots\}, \mathcal{P}(\Omega))$ is equivalent to the specification of:

$(p_n)_n \subset \mathbb{R}^+$ such that $\sum_n p_n = 1$

Correction

Let $\Omega = \{e_1, e_2, \ldots, e_n, \ldots\}$ be a countable sample space, and $\mathcal{P}(\Omega)$ be the power set (the set of all subsets of Ω).

Direction 1: From probability measure to sequence (p_n)

Given a probability measure P on $(\Omega, \mathcal{P}(\Omega))$, define for each n:

$$p_n = P(\{e_n\})$$

Since P is a probability measure, it satisfies:

1. $P(\{e_n\}) \geq 0$ for all n (non-negativity)
2. $P(\Omega) = 1$

Now, since $\Omega = \bigcup_{n=1}^{\infty}\{e_n\}$ is a countable union of disjoint sets, by countable additivity:

$$P(\Omega) = \sum_{n=1}^{\infty} P(\{e_n\}) = \sum_{n=1}^{\infty} p_n = 1$$

Thus, (p_n) is a sequence of non-negative real numbers with $\sum_{n=1}^{\infty} p_n = 1$.

Direction 2: From sequence (p_n) **to probability measure**

Given a sequence $(p_n) \subset \mathbb{R}^+$ such that $\sum_{n=1}^{\infty} p_n = 1$, define a function $P : \mathcal{P}(\Omega) \to [0, 1]$ by:

$$P(A) = \sum_{e_n \in A} p_n \quad \text{for any } A \subseteq \Omega$$

We need to verify that P defined this way is indeed a probability measure.

Verification of probability axioms:

1. **Non-negativity:** For any $A \subseteq \Omega$,

$$P(A) = \sum_{e_n \in A} p_n \geq 0 \quad \text{since } p_n \geq 0 \text{ for all } n$$

2. **Probability of sample space:**

$$P(\Omega) = \sum_{e_n \in \Omega} p_n = \sum_{n=1}^{\infty} p_n = 1$$

3. **Countable additivity:** Let $A_1, A_2, \ldots$ be pairwise disjoint subsets of Ω. Then:

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{e_n \in \bigcup_{i=1}^{\infty} A_i} p_n$$

Since the A_i are disjoint, each elementary outcome e_n belongs to exactly one A_i, so we can rearrange the sum:

$$\sum_{e_n \in \bigcup_{i=1}^{\infty} A_i} p_n = \sum_{i=1}^{\infty} \sum_{e_n \in A_i} p_n = \sum_{i=1}^{\infty} P(A_i)$$

The rearrangement is justified because all terms are non-negative.

Thus, P defined by $P(A) = \sum_{e_n \in A} p_n$ satisfies all the axioms of a probability measure.

Exercise 7:

Study the drawing of N elements from among M distinct elements.

Correction

We consider four cases based on:

- Ordered/Unordered selection
- With/without replacement

1. **Ordered, with replacement:**

$$\text{Number of ways} = M^N$$

Each of the N draws has M choices.

2. **Ordered, without replacement:**

$$\text{Number of ways} = \frac{M!}{(M-N)!} = A_M^N$$

Choose and arrange N distinct elements from M.

3. **Unordered, without replacement:**

$$\text{Number of ways} = \binom{M}{N} = C_M^N = \frac{M!}{N!(M-N)!}$$

Choose N elements from M without regard to order.

4. **Unordered, with replacement:**

$$\text{Number of ways} = \binom{M+N-1}{N}$$

Stars and bars method: number of non-negative integer solutions to $x_1 + \cdots + x_M = N$.

Constraints: For without replacement cases, we require $N \leq M$.

Exercise 8:

We study over time the operation of a device obeying the following rules:

- If the device is working at time $n-1$, ($n \in \mathbb{N}^*$), it has a probability of $\frac{1}{6}$ of being broken at time n.
- If the device is broken at time $n-1$, it has a probability of $\frac{2}{3}$ of being broken at time n.

Let p_n denote the probability that the device is in working order at time n.

1. Establish for all $n \in \mathbb{N}^*$ a relation between p_n and p_{n-1}.
2. Express p_n as a function of p_0.
3. Study the convergence of the sequence (p_n).

Correction

Let p_n be the probability that the device is working at time n.

1) Relation between p_n and p_{n-1}

Using total probability:

$$p_n = P(n \mid n-1) \cdot p_{n-1} + P(n \mid n-1) \cdot (1 - p_{n-1})$$

Given:

- If working at $n-1$: probability $\frac{1}{6}$ of being broken at $n \implies$ probability $\frac{5}{6}$ of working at n
- If broken at $n-1$: probability $\frac{2}{3}$ of being broken at $n \implies$ probability $\frac{1}{3}$ of working at n

So:

$$p_n = \frac{1}{3} + \frac{1}{2} p_{n-1}$$

2) Express p_n as a function of p_0

This is a linear recurrence: $p_n = \frac{1}{2}p_{n-1} + \frac{1}{3}$

Let L be the fixed point: $L = \frac{1}{2}L + \frac{1}{3} \implies L = \frac{2}{3}$

Then $p_n - \frac{2}{3} = \frac{1}{2}(p_{n-1} - \frac{2}{3})$

So:

$$p_n - \frac{2}{3} = \left(\frac{1}{2}\right)^n (p_0 - \frac{2}{3})$$

$$p_n = \frac{2}{3} + \left(\frac{1}{2}\right)^n (p_0 - \frac{2}{3})$$

3) Convergence of (p_n)

Since $\left|\frac{1}{2}\right| < 1$, we have:

$$\lim_{n\to\infty} p_n = \frac{2}{3}$$

The sequence converges to $\frac{2}{3}$ regardless of the initial state p_0.

Exercise 9:

A box contains two balls: one black and one red. We draw n times a ball from this box, replacing it after noting its color. Let A_n be the event "we obtain balls of both colors during the n draws" and B_n be the event "we obtain at most one black ball".

1. Calculate $P(A_n)$ and $P(B_n)$.
2. Are A_n and B_n independent if $n = 2$?
3. Same question if $n = 3$.

Correction

1) Calculate $P(A_n)$ and $P(B_n)$
A_n = "obtain balls of both colors during n draws"
Complement: all same color:

$$P(A_n) = 1 - P(\text{all black}) - P(\text{all red}) = 1 - \left(\frac{1}{2}\right)^n - \left(\frac{1}{2}\right)^n$$

$$P(A_n) = 1 - 2\left(\frac{1}{2}\right)^n = 1 - \left(\frac{1}{2}\right)^{n-1}$$

B_n = "obtain at most one black ball"
Cases: 0 black or 1 black:

$$P(B_n) = P(0 \text{ black}) + P(1 \text{ black}) = \left(\frac{1}{2}\right)^n + n\left(\frac{1}{2}\right)^n$$

$$P(B_n) = (n+1)\left(\frac{1}{2}\right)^n$$

2) Independence for $n = 2$
For $n = 2$:

$$P(A_2) = 1 - \left(\frac{1}{2}\right)^1 = \frac{1}{2}$$

$$P(B_2) = (2+1)\left(\frac{1}{2}\right)^2 = \frac{3}{4}$$

Sample space for 2 draws: BB, BR, RB, RR
A_2 = BR, RB $\implies P(A_2) = \frac{1}{2}$ B_2 = BR, RB, RR $\implies P(B_2) = \frac{3}{4}$ $A_2 \cap B_2$ = BR, RB $\implies P(A_2 \cap B_2) = \frac{1}{2}$
Check independence:

$$P(A_2 \cap B_2) = \frac{1}{2} \quad \text{vs} \quad P(A_2)P(B_2) = \frac{1}{2} \times \frac{3}{4} = \frac{3}{8}$$

Not equal $\implies$ NOT independent.
3) Independence for $n = 3$
For $n = 3$:

$$P(A_3) = 1 - \left(\frac{1}{2}\right)^2 = \frac{3}{4}$$

$$P(B_3) = (3+1)\left(\frac{1}{2}\right)^3 = \frac{1}{2}$$

Sample space: 8 equally likely outcomes A_3 = all except BBB and RRR $\implies$ 6 outcomes $\implies P(A_3) = \frac{3}{4}$ B_3 = RRR, BRR, RBR, RRB $\implies$ 4 outcomes $\implies P(B_3) = \frac{1}{2}$ $A_3 \cap B_3$ = BRR, RBR, RRB $\implies$ 3 outcomes $\implies P(A_3 \cap B_3) = \frac{3}{8}$
Check:

$$P(A_3)P(B_3) = \frac{3}{4} \times \frac{1}{2} = \frac{3}{8} = P(A_3 \cap B_3)$$

Equal $\implies$ YES, independent for $n = 3$.